

When Does Intermediate Reward Actually Help?

Outcome-only credit assignment vs. explicit progress supervision
for multi-turn LLM search agents — Wave-1 Results Report

Xiaoxuan Lei

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Abstract

Reinforcement learning for multi-turn LLM agents almost always optimizes a *sparse outcome reward*: one scalar at the end of a trajectory. Whether an agent benefits from an additional *intermediate reward* — supplied by an explicit reward model, a privileged verifiable heuristic, or recovered implicitly by the credit-assignment (CA) machinery of the RL algorithm itself — remains unresolved, and published comparisons rarely control for reward density, compute, or seeds. We ran a controlled study on the Search-R1 question-answering task: a Qwen3-1.7B agent interleaving reasoning, search calls, and a final answer over up to 4 turns (8 in a stress test). Seven credit-assignment arms — token/turn critics, group-relative estimators (GRPO, GiGPO, HCAPO), a privileged answer-exposure step reward (B1), and its density-matched shuffled control (B1-shuffle) — ran to a fixed 500-step budget across 17 seeded runs (≈ 4.7 k GPU-hours). **Five findings.** (1) *Outcome-only group-relative CA wins and is cheaper*: GiGPO 0.397 and token-GRPO 0.394 final val macro-EM beat every critic and step-reward arm by 4–6 points, using *no* critic network. (2) *Explicit progress content adds nothing over its shuffle* (3 seeds each: B1-shuffle 0.353 $\geq$ B1 0.347 $>$ B0 0.332) — a *reward-density* effect, not supervision. (3) A reproducible *late-training truncation instability* tracks aggressive per-turn credit: HCAPO’s per-turn amplifier collapses (clip $\rightarrow$ 0.75, no EM gain), every critic arm drifts, and everything group-relative-and-flat stays clean. (4) A Monte-Carlo *credit-alignment diagnostic* shows the turn critic’s one-step value delta is uncorrelated with true continuation-value progress (pooled Spearman $\approx$ 0), so the pre-registered λ -bootstrapping sweep was dropped as a reported negative. (5) At an *8-turn horizon* the longer budget lifts the sparse baseline (B0 0.332 $\rightarrow$ 0.368) but the privileged content still does not help (8-turn B1 0.343 $<$ 8-turn B0 0.368). Together these close all three of the CA survey’s systemic reporting gaps (GPU-hours, a compute-controlled baseline, and variance) in one study.

1 Motivation and hypotheses

Agentic RL fine-tuning of LLMs is bottlenecked by **credit assignment**: the outcome reward arrives only after a sequence of interdependent decisions. The community’s responses fall into two families rarely compared under matched conditions: (i) **implicit reward models** — a learned value network (token or turn critic) is functionally a per-step reward model, and group-relative estimators (GRPO, GiGPO, HCAPO) assign per-step credit through grouped baselines and anchor-state matching without a critic; and (ii) **explicit intermediate rewards** — a process reward injected at each step from a PRM, an LLM judge, or a verifiable heuristic. The central question is **when the second family adds value beyond the first**, and whether apparent gains survive a control that matches a reward’s density and magnitude but destroys its timing and content. That control is the study’s key instrument. The pre-registered hypotheses (fixed before any curve was seen; docs/plan.md):

- **H1 (granularity)**. Turn-level credit should not underperform token-level PPO at matched compute.

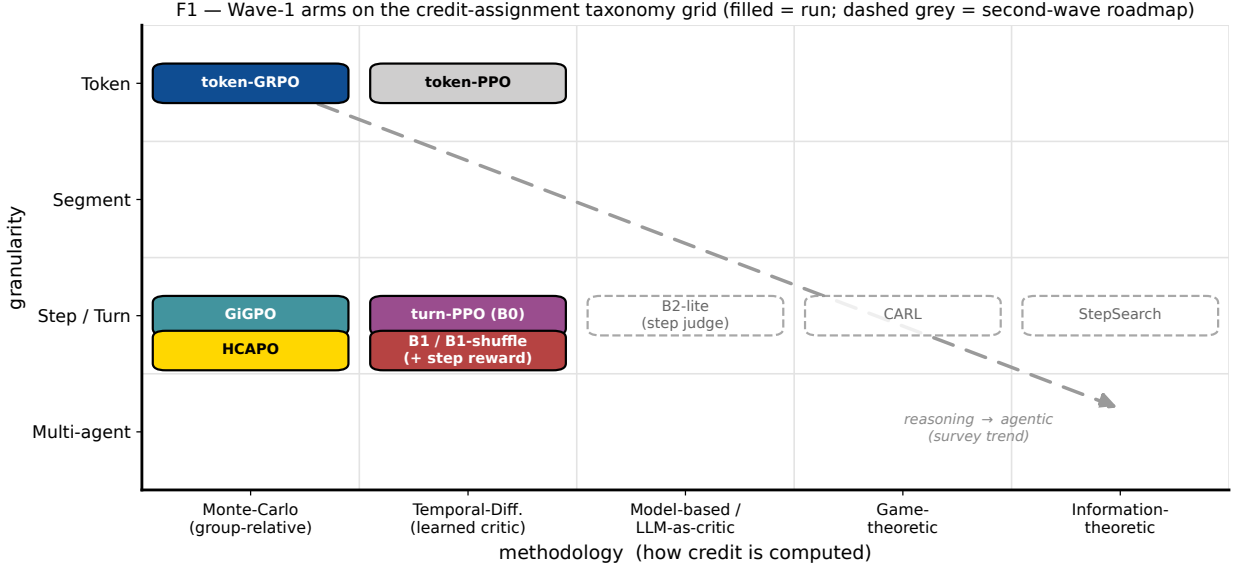


Figure 1: F1 — the Wave-1 arms on the survey’s credit-assignment taxonomy grid (granularity $\times$ methodology, arXiv 2604.09459 Fig. 2). Filled chips are run in Wave 1, coloured by family; dashed grey chips are the second-wave roadmap (B2-lite step judge, CARL, StepSearch). The study populates the Monte-Carlo (group-relative) and Temporal-Difference (learned-critic) columns at Token and Step/Turn granularity — exactly the contrast H1–H3 target.

- **H2 (implicit recovery).** Outcome-only group-relative CA can recover useful intermediate credit — measured by accuracy *and* by a quantitative credit-alignment diagnostic.
- **H3 (content vs. density — core).** If explicit progress supervision helps through its *information content*, B1 must beat the density-matched B1-shuffle. If $B1 \approx B1\text{-shuffle} > B0$, the benefit is a density/optimization effect any dense signal provides.
- **H4 (horizon).** Any content benefit should grow with task horizon (8-turn budget; multi-hop subgroup).
- **H5 (scalable evaluator, staged).** A prompted step evaluator is pursued only if H3 shows content matters.

2 Experimental design

Task. Search-R1 protocol: the agent answers open-domain questions by interleaving `<think>` reasoning with either `<search>query</search>` or a final `<answer>...</answer>`, for up to 4 turns (main) or 8 (stress test). Retrieval is dense e5 over the 2018 Wikipedia dump (wiki-18, 61 GB FAISS index), top-3 passages/query, served by a co-located retrieval server. **Training data:** the standard Search-R1 mixture (NQ + HotpotQA train, HF `PeterJinGo/nq_hotpotqa_train`), versioned preprocessing. **Validation** during training uses a fixed stratified 2,048-question subset (`val_2048`, greedy, every 25 steps); final evaluation is the full paper-protocol test set (**51,713 questions, 7 datasets**, greedy EM). **Reward:** rule-based EM on the final answer plus a small invalid-action penalty — no learned reward model in the core tier, so the only intermediate signal is the one we inject and control. **Model:** Qwen3-1.7B (thinking mode, 2,048-token responses, 4,096-token prompts), identical across arms; batch 256, group 5, 500 steps, matched LR/KL. A pre-registered feasibility gate on the *untrained* model (macro-EM floor ≥ 0.03 , valid-action $\geq 90\%$, search-invocation $\geq 50\%$, within-group reward variance) passed at 1.7B, so the fleet stayed at this scale.

Pre-registered decision rules. Primary comparison = fixed-budget *final* checkpoint (@500); sample-efficiency = validation AUC; best-val checkpoint reported but never the headline. RQ3 is

Arm	Estimator	Critic	Where the “reward model” lives
TOKEN-PPO	GAE (token)	yes	implicit: token-level value network
TURN-PPO/B0	GAE (turn)	yes	implicit: turn-level value network
TOKEN-GRPO	GRPO	no	implicit: group-relative outcome baseline
GiGPO	GiGPO	no	implicit: episode + anchor-state grouped baselines
HCAPO	HCAPO	no	implicit: hindsight-conditioned per-turn advantage amplifier
B1	GAE (turn) + step reward	yes	explicit: privileged answer-exposure signal, +0.2 the first turn whose retrieved passages contain the gold answer
B1-SHUFFLE	GAE (turn) + shuffled step reward	yes	explicit, contentless: B1’s rewards re-assigned to another non-terminal search turn — matched count/magnitude, destroyed timing

Table 1: The seven core arms. B1 is deliberately *privileged and verifiable* (it peeks at the gold answer) — an upper-bound proxy for a perfect progress reward, free of judge noise. The shuffled control isolates information content from reward density.

judged on per-seed final-EM deltas with seed ranges (2–3 seeds/arm) plus a paired per-dataset breakdown with the multi-hop subgroup as the mechanism readout. **Interpretation grid:** B1 > shuffle $\approx$ B0 $\Rightarrow$ content matters; B1 $\approx$ shuffle > B0 $\Rightarrow$ density effect; all $\approx$ $\Rightarrow$ this signal does not help at this horizon. The `shuffle_active_frac` (fraction of B1-positive trajectories with ≥ 2 eligible search turns) is reported beside any B1-vs-shuffle claim; a Wave-1 value below $\sim 30\%$ promotes the 8-turn stress test from optional to *required* (it did — §3.2).

3 Results

All 17 runs reached the 500-step budget. Table 2 and Fig. 2 give the pre-registered primary comparison; Fig. 3 the learning curves.

Arm	Credit family	Gran.	Seeds	Mean	Range	Clip health
GiGPO	critic-free group-rel.	turn	2	0.397	0.396–0.398	clean (≈ 0)
TOKEN-GRPO	critic-free group-rel.	traj.	2	0.394	0.389–0.399	clean (≈ 0)
HCAPO	critic-free amplifier	turn	1	0.357	—	collapse, clip 0.754
B1-SHUFFLE	critic + shuffled step	turn	3	0.353	0.340–0.363	clean
B1	critic + privileged step	turn	3	0.347	0.318–0.372	drift; s1 collapsed
TURN-PPO/B0	critic	turn	3	0.332	0.324–0.343	drift, all seeds
TOKEN-PPO	critic	token	1	0.315	—	persistent drift
B0, 8-turn	critic	turn	1	0.368	—	clean-ish (0.037)
B1, 8-turn	critic + step	turn	1	0.343	—	drift (0.15→0.27)

Table 2: Final val-2048 macro-EM at the fixed-budget checkpoint (@500), the pre-registered primary metric. Untrained baseline 0.2246. Range = min–max across seeds. “Clip” = response-length truncation ratio (fraction of turn responses hitting the 2,048-token cap; healthy ≈ 0). Backing data: `assets/figs/finals.csv`, one row per seeded run, regenerated from the launch logs.

3.1 Finding 1 — outcome-only group-relative CA wins, and costs less

GiGPO (0.397) and token-GRPO (0.394) beat every critic-based and step-reward arm by 4–6 points (Table 2, Fig. 2), solidified at two seeds each with clean clip health throughout. Both use *no* critic

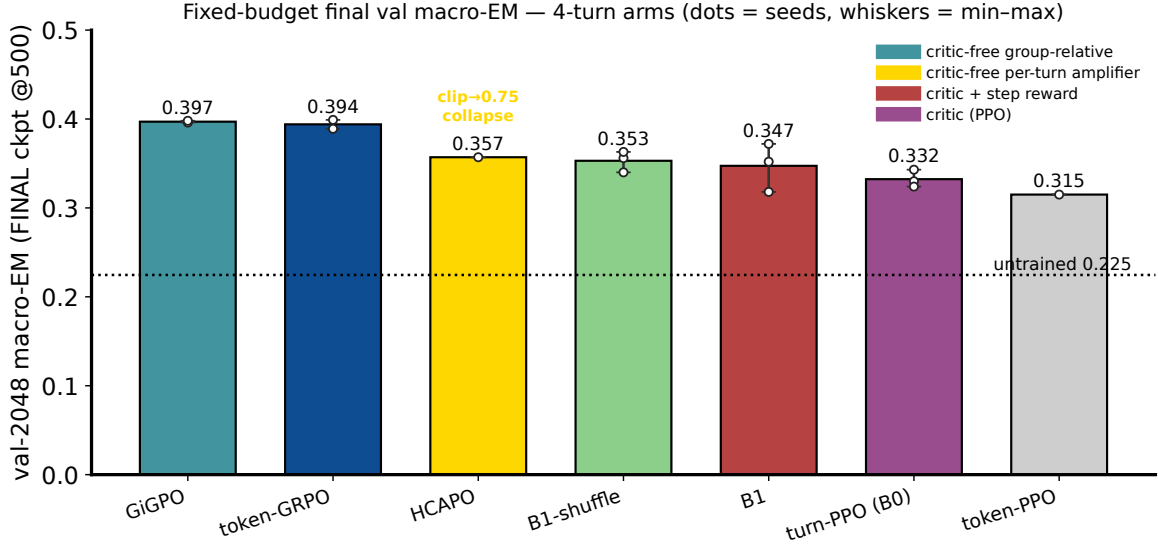


Figure 2: Headline: fixed-budget final val macro-EM, 4-turn arms. Dots are individual seeds, whiskers min-max. Outcome-only group-relative methods (GiGPO, token-GRPO; cool colours) lead by 4–6 points over every critic and step-reward arm — *without* a critic network. HCAPO reaches a middling 0.357 but only via a truncation collapse (clip→0.75, flagged; Fig. 6). The explicit progress reward (B1) does not beat its shuffled placebo (B1-shuffle).

network, and on wall-clock they were the *cheapest* arms (≈ 29 – 30 h vs. ≈ 33 – 37 h for the critic arms on the same $8\times H100$ worker; §4) — the critic is pure overhead here that also hurts accuracy and stability. H1 (granularity) holds directionally: turn-PPO (0.332) $\geq$ token-PPO (0.315) at matched config, though both trail the critic-free arms. This is the study’s cleanest positive result and directly answers H2 on accuracy; Finding 4 supplies the mechanism.

3.2 Finding 2 (RQ3) — privileged content adds nothing over its shuffle

At three seeds per arm the pre-registered comparison reads

$$\text{B1-shuffle } 0.353 \geq \text{B1 } 0.347 > \text{B0 } 0.332,$$

i.e. both dense-reward arms beat sparse B0 by a small margin, but the *shuffled placebo matches or exceeds the real privileged content* (Fig. 4). Per the pre-registered interpretation rule (“B1 $\approx$ B1-shuffle $>$ B0 $\Rightarrow$ density/optimization effect, not supervision”), this is **a reward-density effect, not progress-content supervision**. The dynamics corroborate the reading (Fig. 5): the contentless shuffle arm *saturates the turn budget* (final avg. 4.00 turns) — a dense reward on search turns teaches “search *more*”, not “search *better*” — yet loses no accuracy and stays clip-clean, whereas B1 answers in fewer turns (avg. 3.00) but destabilises (clip 0.19). B1’s seed variance is wide (0.318–0.372); the shuffle is tighter (0.340–0.363), so the control is also the more reliable arm.

On the validity of the 4-turn control. `shuffle_active_frac` was 0.035 on the *base*-model distribution — far below the 0.30 rule — which fired the pre-registered plan B and *promoted the 8-turn stress test to required* (Finding 5). But during RL the metric *inverted*: as the policy learned to issue more searches, `active_frac` rose past 0.30 to ≈ 0.97 – 1.0 on the training batches that produced the finals. So the 4-turn control was in fact *fully discriminating* for the reported comparison — the density reading does not hinge on a degenerate control.

3.3 Finding 3 — truncation instability tracks aggressive per-turn credit

The single most reproducible *mechanistic* result is a late-training response-truncation runaway (Fig. 6). It scales with how aggressively an arm concentrates credit on individual turns: HCAPO

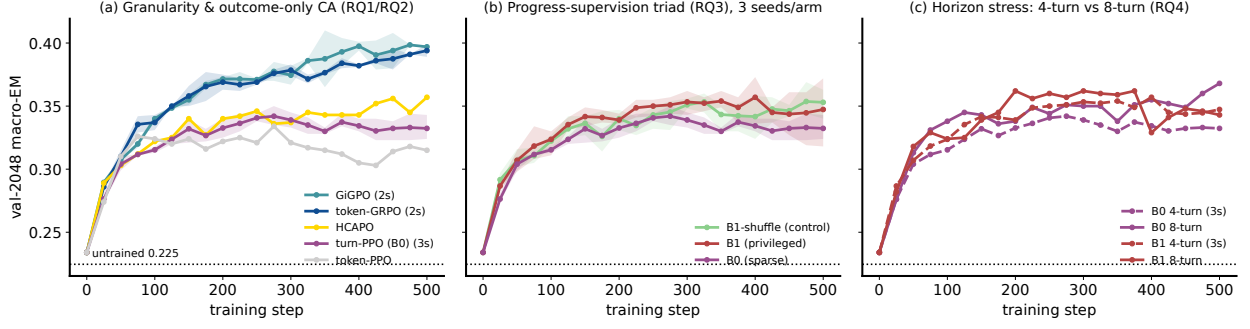


Figure 3: Validation macro-EM (val_2048, greedy, every 25 steps); band = seed min–max. **(a)** Granularity & outcome-only CA: GiGPO and token-GRPO climb steadily to ≈ 0.40 ; the critic PPO arms plateau near 0.33; token-PPO is weakest; HCAPO (amber) plateaus at 0.35 as it destabilises. **(b)** The RQ3 triad at 3 seeds/arm: the three conditions are statistically inseparable throughout, with B1-shuffle (control) tracking at or above B1 (privileged content). **(c)** Horizon stress: the 8-turn budget lifts the sparse B0 baseline above its 4-turn self, but 8-turn B1 sits below 8-turn B0.

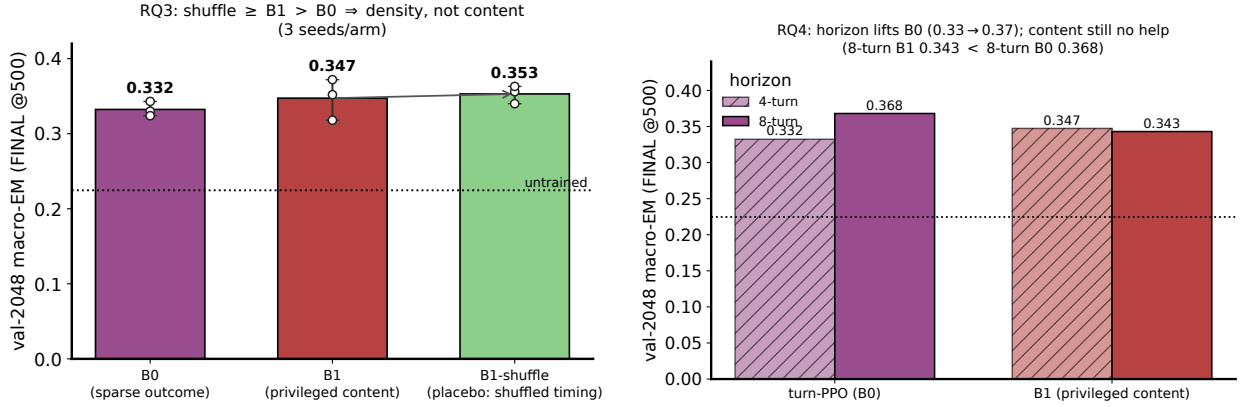


Figure 4: Left (RQ3, Finding 2): the privileged-content arm (B1) does not beat its density-matched placebo (B1-shuffle); both edge out sparse B0. 3 seeds/arm, dots per seed. **Right (RQ4, Finding 5):** the 8-turn horizon lifts the sparse baseline (0.332 $\rightarrow$ 0.368), but the privileged content still does not help (8-turn B1 0.343 < 8-turn B0 0.368); the step-200 early hint (B1 0.358 > B0 0.338) did not hold to the final.

— a critic-free *per-turn advantage amplifier* — blows up completely (clip $\rightarrow$ 0.754, generating over-long per-turn responses that hit the 2,048-token cap 75% of the time) while its accuracy merely plateaus at ≈ 0.35 , so the “collapse” buys nothing; every critic arm drifts upward (token-PPO 0.10, B0 0.14, B1 0.19, 8-turn B1 0.27); and everything group-relative-and-flat (GiGPO, token-GRPO, and the shuffled B1) stays at clip ≈ 0 throughout. This reframes late-training length instability as a *first-class cost* of aggressive per-turn credit weighting — a stability, not just accuracy, axis on which the outcome-only group-relative methods dominate. It is also the mechanism behind B1’s wide seed variance (its s1 collapsed to 0.318).

3.4 Finding 4 (F8a) — the critic assigns credit uncorrelated with true progress

To test H2 quantitatively — does a method assign *correct* intermediate credit, not just train a better policy? — we ran the pre-registered credit-alignment diagnostic on the turn critic. For logged trajectories at fixed B0 checkpoints we estimated each prefix’s outcome-only Monte-Carlo continuation value $\hat{V}(s_t) = \frac{1}{K} \sum_k R(\tau_k | s_t)$ by resuming from the prefix under the *same* checkpoint ($K=8$), giving the martingale-increment $\Delta \hat{V}_t = \hat{V}(s_{t+1}) - \hat{V}(s_t)$ — the credit an ideal process reward would assign. We compared it to the critic’s own one-step delta $A_t = V_\phi(s_{t+1}) - V_\phi(s_t)$. Across all four checkpoints the pooled Spearman correlation is indistinguishable from zero ($\rho \in$

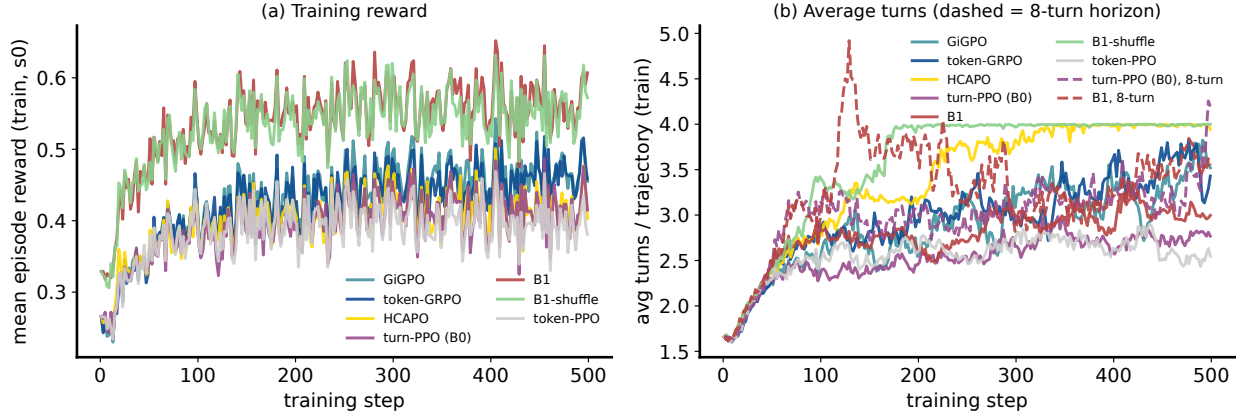


Figure 5: Training dynamics (seed-0 arms). **(a)** Mean episode reward; B1/B1-shuffle sit higher by construction (the +0.2 step bonus). **(b)** Average turns per trajectory. Note B1-shuffle saturating the 4-turn budget and HCAPO’s high turn usage; dashed = 8-turn arms, which use only ≈ 4.4 of 8 available turns.

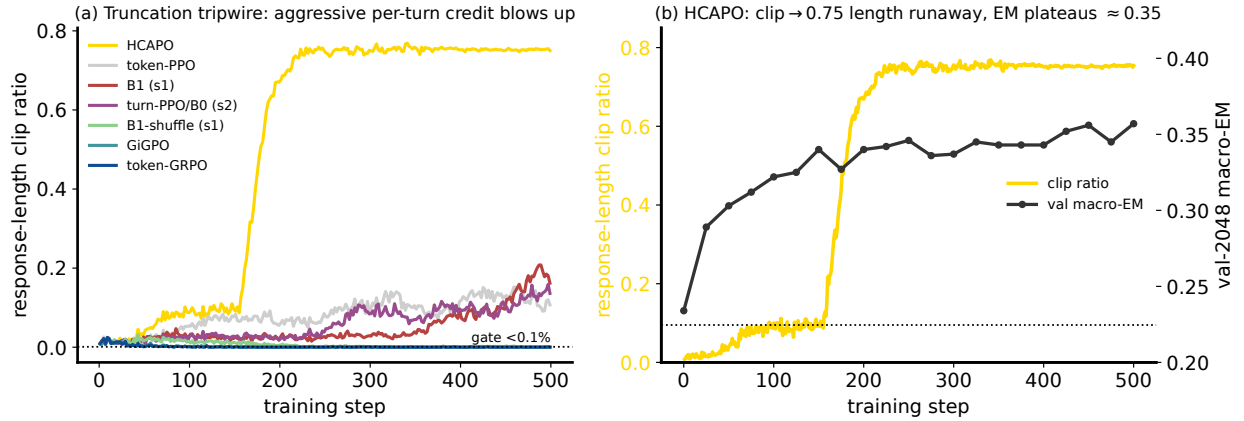


Figure 6: Finding 3 — the truncation tripwire. **(a)** Response-length clip ratio over training (must fall to ≈ 0). HCAPO’s per-turn amplifier runs away to 0.75; the critic arms drift up; group-relative-and-flat arms (GiGPO, GRPO, B1-shuffle) stay clean. **(b)** HCAPO in detail: clip climbs to 0.75 while val macro-EM plateaus at ≈ 0.35 — the length runaway yields no accuracy, it is pure degeneration.

$[-0.02, 0.11]$, bootstrap 95% CIs spanning 0; sign agreement ≈ 0.15 ; Fig. 7), so **0 of 3 gate checkpoints passed the pre-registered unlock threshold** ($\rho > 0.2$ with CI excluding 0) and the planned λ -bootstrapping sweep was **dropped and is reported here as a negative**. The turn critic functions only as a weak trajectory-level baseline (value explained-variance ≈ 0.28), *not* as a per-turn reward model. This is the mechanism behind Finding 1: bootstrapping through a critic that carries essentially no per-turn progress information ($\lambda < 1$) has no basis here, which is exactly why the critic arms trail the flat group-relative baselines.

3.5 Finding 5 (RQ4) — horizon lifts the baseline, content still does not help

The 8-turn stress test (promoted to required by the base-distribution active_frac; §3.2) gives a decisive horizon readout (Fig. 4, right). The longer budget *lifts the sparse baseline itself*: 8-turn B0 = 0.368 $>$ 4-turn B0 mean 0.332. But the privileged progress content still does not help: **8-turn B1 = 0.343 $<$ 8-turn B0 = 0.368**, and the step-200 early hint (B1 0.358 $>$ B0 0.338) did *not* hold to the final — B1’s added reward density drove clip drift (0.15 $\rightarrow$ 0.27) and a mild degradation (peak 0.362 $\rightarrow$ 0.343). So the marginal value of this progress supervision does *not* grow with horizon; it is ≤ 0 here, consistent with the 4-turn density reading. *Caveat (pre-logged)*: no 8-turn shuffle control was run, so 8-turn B1 $<$ B0 cannot fully separate content from density — but the direction (no benefit, mild harm) is unambiguous and matches the fully-controlled 4-turn

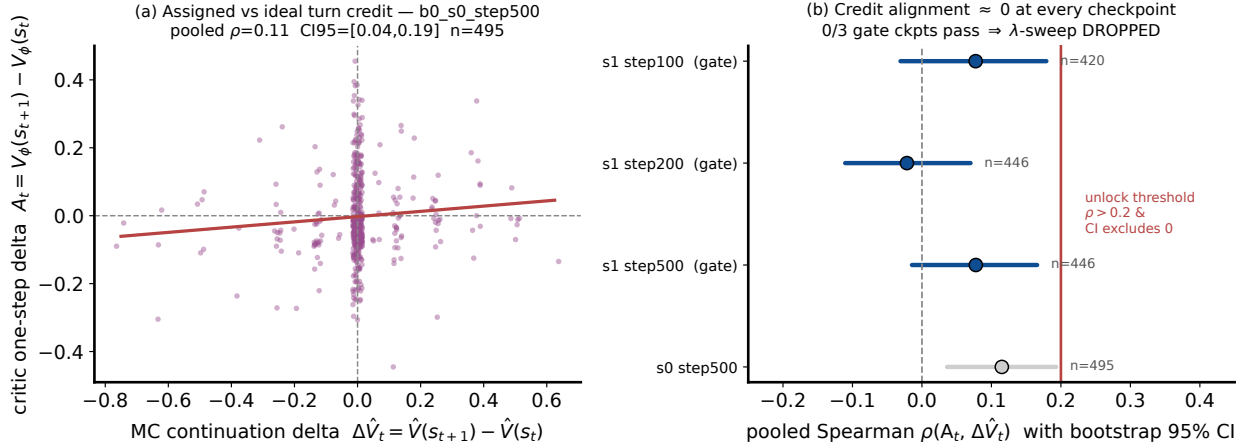


Figure 7: Finding 4 — credit alignment ≈ 0 . (a) Critic one-step delta A_t vs. the Monte-Carlo continuation delta $\Delta\hat{V}_t$ (the “ideal” credit), pooled over turn-pairs at the best-populated checkpoint ($n=495$); the least-squares fit is nearly flat. (b) Pooled Spearman with bootstrap 95% CI at every diagnostic checkpoint; none clears the $\rho > 0.2$ unlock threshold, so the λ -sweep is dropped as a pre-registered negative.

result.

3.6 Full-set evaluation (proxy validation) and the multi-hop subgroup

One arm has completed the full 51,713-question paper-protocol evaluation so far — token-GRPO seed 0 at its final checkpoint (Table 3). Its full-set macro-EM is **0.3951** (micro 0.4433), within ≈ 0.006 of its val_2048 proxy (0.389) — **validating the fixed subset as the training-time metric**. The pre-registered single- vs multi-hop split is stark: 0.499 single-hop vs. 0.317 multi-hop, with MuSiQue (the hardest 4-hop set) at 0.142. Turn usage rises with difficulty (avg. 3.1 single-hop $\rightarrow$ 3.6 MuSiQue) and *no* trajectory was left unterminated (0% truncation on the answer turn). Full-set evals of the remaining final checkpoints are the immediate Wave-4 GPU task; they will populate the per-dataset Δ EM breakdown for every arm.

Dataset	Subgroup	n	EM	avg. turns	unterm.
NQ	single-hop	3,610	0.445	3.15	0%
TriviaQA	single-hop	11,313	0.602	3.12	0%
PopQA	single-hop	14,267	0.450	3.14	0%
HotpotQA	multi-hop	7,405	0.396	3.31	0%
2WikiMultihopQA	multi-hop	12,576	0.378	3.46	0%
MuSiQue	multi-hop	2,417	0.142	3.61	0%
Bamboogle	multi-hop	125	0.352	3.58	0%
macro / micro		51,713	0.395 / 0.443	3.3	0%
single-hop / multi-hop macro			0.499 / 0.317		

Table 3: Full-set paper-protocol EM for token-GRPO s0 (final checkpoint). Base untrained model on the same protocol: 0.301 single-hop / 0.167 multi-hop macro. Source: eval_full/wave0_grpo_s0_final/paper_table.json.

4 Compute accounting

Closing the survey’s #1 reporting gap (0/41 CA papers report GPU-hours): the 17 completed 500-step training runs used **$\approx 4.7\text{k GPU-hours}$** ($8 \times \text{H100}$ each; per-run wall-clock measured launch-to-completion on clean runs — $\approx 29\text{--}30\text{ h}$ for a 4-turn critic-free run, $\approx 33\text{--}37\text{ h}$ for a 4-turn critic run, $\approx 45\text{ h}$ for an 8-turn run, $\times 8$ GPUs). Including the failed-attempt reruns (early-Wave OOM

cycles resolved by a micro-batch protocol fix), the F8a diagnostic worker sessions, the full-set eval, and the Wave-0 feasibility/toy-gate runs, the project total is $\approx 5.5\text{k GPU-hours}$. **CA overhead by arm** (survey item #12): critic-free group-relative CA has *negative* overhead — it removes the critic’s forward/backward, running $\approx 15\text{--}20\%$ faster per step than PPO; HCAPO adds a hindsight relabelling pass (cheap in compute, expensive in stability); B1/B1-shuffle add only an env-side gold-answer string match per turn (and a seeded driver-side reassignment for the shuffle). The expensive add-on is the offline F8a diagnostic (prefix-resume + K continuations per prefix), one worker for a few hours.

5 Discussion

The five findings converge on one thesis: **for this multi-turn search task, outcome-only group-relative credit assignment is the right default** — it is the most accurate, the most stable, and the cheapest, and explicit privileged progress supervision adds no value that its own density-matched shuffle does not already provide, at either 4 or 8 turns. The critic-based alternative is not merely weaker but *mechanistically* deficient: its per-turn value deltas carry essentially no true-progress information (Finding 4), which explains both its accuracy gap and — through aggressive per-turn credit weighting — its truncation instability (Finding 3).

Scope of the claim (from the pre-registration, not hedging after the fact). This is evidence about *this* privileged answer-exposure signal at *this* horizon on *this* task; it is *not* a claim that progress reward models are useless in general. B1 is a verifiable upper-bound proxy free of judge noise — that it fails to beat its shuffle is a strong negative for the “dense progress signal helps” hypothesis in this regime, but a genuinely informative learned PRM on a longer-horizon task with sparser reward could still help; that is the B2/CARL second wave. **Open items:** full-set evals of the remaining final checkpoints (the per-dataset ΔEM breakdown; Wave-4 GPU); HCAPO/token-PPO second seeds; and the second-wave methods (CARL, B2-lite step judge, StepSearch) each behind their own gate.

A Reporting checklist (survey Table 11) — filled

Per the project standard, every report fills the CA survey’s reporting checklist (docs/CREDIT_ASSIGNMENT_CHECKLIST.md). This study was *designed* to close the survey’s three systemic gaps (GPU-hours 0/41; compute-controlled baseline 0/41; variance 2/41).

#	Category	Priority	How this study reports it
1	Model & Data	Req.	Qwen3-1.7B, thinking mode, HF revision pinned in configs
2	Model & Data	Req.	Search-R1 NQ+HotpotQA train; versioned preprocessing; 51,713-q / 7-dataset test
3	CA Method	Req.	granularity per arm (token / turn / trajectory), Table 2 + F1
4	CA Method	Req.	methodology family per arm (MC group-relative / TD critic), F1 grid
5	Baselines	Req.	episode-level token-PPO <i>and</i> token-GRPO, identical base model
6	Baselines	Req.	compute-controlled : B1-shuffle is density-matched; iso-worker critic-vs-critic-free wall-clock reported
7	Baselines	Rec.	CA-component isolation: B1 vs B1-shuffle (content vs density); λ -gate on the critic
8	Evaluation	Req.	val_2048 (train-time) + full 51,713-q paper test, exact splits named
9	Evaluation	Req.	macro/micro EM, exact match on final answer, defined in §3
10	Evaluation	Rec.	variance : 2–3 seeds/arm, min–max ranges in Table 2
11	Compute	Req.	GPU-hours : $\approx 4.7k$ training / $\approx 5.5k$ total, §4
12	Compute	Rec.	CA overhead per arm (critic-free negative; HCAPO hindsight; diagnostic cost), §4
13	Trajectory	Req.	avg. turns per arm (Table 2 data; 2.5–4.0), full-set avg. 3.3
14	Trajectory	Rec.	turns min/median/max per dataset, Table 3 + paper_table.json

Table-12 scorecard (this study vs. the survey’s three exemplars). ✓ reported, ~ partial, ✗ not reported.

Category	Item	HICRA	GiGPO	M-GRPO	This study
Model	Base model name/size	✓	✓	✓	✓
Model	Training data source/size	~	~	✗	✓
CA Method	Credit granularity	✓	✓	✓	✓
CA Method	Methodology family	✓	✓	✓	✓
Baselines	Episode-level baseline (same model)	✓	✓	✓	✓
Baselines	Compute-controlled baseline	✗	✗	✗	✓
Baselines	CA-component ablation	✓	~	✗	✓
Evaluation	Benchmark names + splits	✓	✓	~	✓
Evaluation	Variance / CIs	~	✗	✗	✓
Compute	Total GPU-hours	✗	✗	✗	✓
Compute	CA-specific overhead	~	~	✗	✓
Trajectory	Avg trajectory length	~	✓	✗	✓

Table 4: The study reports all three of the survey’s systemic gaps (bold rows): compute-controlled baseline, variance across seeds, and total GPU-hours.

B Annotated example trajectory and per-method credit

A trained token-GRPO trajectory from the full-set eval (traj a1c80bb0, HotpotQA, 3 turns, EM = 1; reconstructed from the logged rollout). Turn-by-turn text with the credit each family *would* assign, so the credit assignment is inspectable:

QUESTION: Are Scott Derrickson and Ed Wood of the same nationality?	
TURN 0	<think>I need each director's nationality. Start with Ed Wood.</think> <search>Ed Wood nationality</search> <information>Doc: Edward D. Wood Jr. was an American filmmaker ...</information>
TURN 1	<think>Ed Wood is American. Now Scott Derrickson.</think> <search>Scott Derrickson nationality</search> <information>Doc: Scott Derrickson is an American director (Doctor Strange) ...</information>
TURN 2	<think>Both are American, so the answer is yes.</think> <answer>yes</answer> --> outcome reward 1.0

What each arm assigns to this trajectory (turns 0,1,2):

- TOKEN-GRPO: one trajectory-level advantage (group-relative to the batch), broadcast uniformly to every response token of all three turns — no per-turn distinction.
- GiGPO: a group-relative advantage *per turn*, comparing each turn against an anchor group of same-state siblings — turns 0 and 1 (both answer-advancing searches) score positively.
- TURN-PPO/B0: the critic’s per-turn value delta A_t — which Finding 4 shows is essentially uncorrelated with the true continuation-value progress of turns 0→1→2.
- B1: B0’s credit *plus* +0.2 on turn 0 (the first retrieval containing the gold string “American”), first-hit-only.
- B1-SHUFFLE: the same +0.2 re-assigned to another eligible non-terminal search turn (here turn 1) — matched magnitude, destroyed timing. Findings 2/5: this changes nothing.

Reproducibility. Figures + finals: `assets/extract_metrics.py` (launch-log → per-run CSV) → `assets/build_figures.py` + `build_f8a.py` + `build_taxonomy.py` (see `assets/figs/README.md` for the per-figure data map). Training logs, the worker ledger (GPU-hours), and the append-only decision log live in the project repository (**Agentic-RL-CA**, branch `agentic-rl-ca`); pre-registration in `docs/plan.md`; the credit-alignment diagnostic in `credit_assignment/{diagnostic,lambda_gate}.py`.